BACHELOR OF COMPUTER APPLICATION

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 10

Differentiation of Trigonometric and Inverse Circular Function

Table of Contents

SL.No	Topic		Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction		-	-	3 - 4
	1.1	Learning Objectives	-	-	
2	How to Differentiate a Function		-	-	5 - 10
	2.1	Derivative of Trigonometric Functions	-	-	
	2.2	Derivative of Inverse Trigonometric Functions	-	-	
	2.3	Derivative of Exponential Functions	-	-	
	2.4	Derivative of Logarithmic Functions	-	-	
	2.5	Derivative of Hyperbolic Functions	-	-	
	2.6	Derivative of Inverse Hyperbolic Functions	-	-	
3	Working Examples		-	-	11 – 13
4	Summary		-	-	14 - 15
5	Self-Assessment Questions		-	1	16
6	Terminal Questions		-	-	17
7	Answers		-	-	18

1. INTRODUCTION

Welcome to Unit 10: Differentiation of Trigonometric Functions and Inverse Circular Functions. In Unit 9, we explored the foundations of differentiation, including key theorems like the Chain Rule, Product Rule, and Quotient Rule, which equip you with the ability to handle various complex functions smoothly. We also delved into implicit differentiation and higher-order derivatives, strengthening your calculus toolkit to analyse the dynamic behavior of functions with precision.

In Unit 10, we turn our focus to a specific yet profoundly impactful area of calculus—differentiation of trigonometric and inverse circular functions. This unit is not just about learning to differentiate these functions; it's about understanding their significance in real-world applications. From physics to engineering, the derivatives of sine, cosine, tangent, and their inverse functions play crucial roles in modelling periodic phenomena, oscillations, and wave patterns. By mastering these concepts, you'll be able to tackle challenges in numerous scientific fields, enhancing your analytical skills and problem-solving capabilities.

To get the most out of this unit, I recommend a layered approach to learning. Start by revisiting the basic properties and graphs of trigonometric functions if you need a refresher. As you progress through the lessons, try to link each derivative with its geometric interpretation—this will help in visualizing the concepts and understanding their practical implications. Practice regularly with the exercises provided, and don't hesitate to use simulation tools to see these derivatives in action. Engage with the material actively by forming study groups, discussing with peers, and using online forums to clarify doubts. Let's dive into the intricate dance of angles and rates, where each new formula you learn is a step towards mastering the art of calculus.

1.1. Learning Objectives

At the end of this unit, you will be able to:

- *Recall the formulas for the derivatives of basic trigonometric functions (sine, cosine, tangent).*
- *Explain the significance of the derivatives of trigonometric functions in real-world contexts.*
- *Calculate the derivatives of given trigonometric functions using differentiation rules.*
- *Differentiate between the processes of finding derivatives for trigonometric functions and their inverses.*

2. HOW TO DIFFERENTIATE A FUNCTION

Differentiation is a fundamental concept in calculus that involves finding the rate at which a function changes at any given point. This guide covers the differentiation of various types of functions, including trigonometric, inverse trigonometric, exponential, logarithmic, hyperbolic, and inverse hyperbolic functions.

2.1. Derivative of Trigonometric Functions

Trigonometric functions are fundamental in mathematics, especially in geometry and periodic phenomena. Their derivatives are essential in various applications, such as calculating the motion of waves, oscillations, and circular motion.:

1. **Sine Function**

   The derivative of the sine function is the cosine function, reflecting how the rate of change of sine varies with its argument.

$$\frac{d}{dx} sin\, x = cos\, x$$

2. **Cosine Function**

   The derivative of the cosine function is the negative sine function, indicating that the rate of change of cosine is out of phase with sine by 90 degrees.

$$\frac{d}{dx} \cos x = -\sin x$$

3. **Tangent Function**

   The derivative of the tangent function is the square of the secant function, highlighting the rapid change in the slope of the tangent function as it approaches its vertical asymptotes.

$$\frac{d}{dx} \tan x = \sec^2 x$$

4. **Cosecant Function**

   The derivative of the cosecant function involves both the cosecant and cotangent functions, reflecting their interdependence.

$$\frac{d}{dx} \operatorname{cosec} x = -co \sec x \cdot \tan x$$

5. **Secant Function**

   The derivative of the secant function involves both the secant and tangent functions, showing how the secant function's slope changes with its argument.

$$\frac{d}{dx}\sec x = \sec x \cdot \tan x$$

6. **Cotangent Function**

   The derivative of the cotangent function is the negative square of the cosecant function, similar to how the tangent function's derivative involves the secant function.

$$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$$

2.2. Derivative of Inverse Trigonometric Functions

Inverse trigonometric functions, which provide angles corresponding to given trigonometric values, have specific derivatives that are useful in various applications, such as solving trigonometric equations and integration.

1. **Arcsine Function**

   The derivative of the arcsine function shows how the rate of change of the angle varies with its sine value.

$$\frac{d}{dx} sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

2. **Arccosine Function**

   The derivative of the arccosine function is the negative of the arcsine derivative, reflecting their complementary nature.

$$\frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1-x}^2}$$

3. **Arctangent Function**

   The derivative of the arctangent function indicates how the rate of change of the angle varies with its tangent value.

$$\frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}$$

4. **Arccosecant Function**

   The derivative of the arccosecant function involves the absolute value of the argument, reflecting its domain restrictions.

$$\frac{d}{dx}\operatorname{cosec}^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$$

5. **Arcsecant Function**

   The derivative of the arcsecant function is similar to the arccosecant derivative but without the negative sign.

$$\frac{d}{dx}\sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$$

6. **Arccotangent Function**

   The derivative of the arccotangent function is the negative of the arctangent derivative, reflecting their complementary nature.

$$\frac{d}{dx}\cot^{-1}(x) = -\frac{1}{1+x^2}$$

2.3. Derivative of Exponential Functions

Exponential functions, which describe growth and decay processes, have derivatives that reflect their rapid change. These functions are essential in modelling natural phenomena such as population growth, radioactive decay, and continuously compounded interest.

1. **Natural Exponential Function**

   The derivative of the natural exponential function is unique because it is the same as the original function, highlighting the constant rate of change relative to its value.

$$\frac{d}{dx}e^x = e^x$$

2. **General Exponential Function**

   The derivative of a general exponential function involves the natural logarithm of the base, reflecting the base's influence on the rate of change.

$$\frac{d}{dx} = a^x \ln(a)$$

2.4. Derivative of Logarithmic Functions

Logarithmic functions, the inverses of exponential functions, have derivatives that reveal the rate of change relative to their argument. They are widely used in applications involving multiplicative processes and logarithmic scales, such as pH levels, sound intensity, and earthquake magnitudes.

1. **Natural Logarithm**

   The derivative of the natural logarithm function shows how the rate of change decreases as the argument increases, reflecting the logarithmic scale.

$$\frac{d}{dx} ln(x) = \frac{1}{x}$$

2. **Common Logarithm**

   The derivative of the common logarithm function involves the natural logarithm of the base, indicating how the base affects the rate of change.

$$\frac{d}{dx} \log_a x = \frac{1}{x \ln(a)}$$

2.5. Derivative of Hyperbolic Functions

Hyperbolic functions, analogous to trigonometric functions but based on hyperbolas, have unique derivatives that are useful in various mathematical and physical applications, such as solving differential equations and modelling hyperbolic geometry.

1. **Hyperbolic sine**

   The derivative of the hyperbolic sine function is the hyperbolic cosine function, similar to the relationship between the sine and cosine functions.

$$\frac{d}{dx} sin\, h(x) = cosh(x)$$

2. **Hyperbolic cosine**

   The derivative of the hyperbolic cosine function is the hyperbolic sine function, indicating their interdependence.

$$\frac{d}{dx} \cosh(x) = \sin h(x)$$

3. **Hyperbolic Tangent**

   The derivative of the hyperbolic tangent function is the square of the hyperbolic secant function, similar to the derivative of the tangent function.

$$\frac{d}{dx} \tanh(x) = \operatorname{sech}^2(x)$$

4. **Hyperbolic cosecant**

   The derivative of the hyperbolic cosecant function involves both the hyperbolic cosecant and cotangent functions.

$$\frac{d}{dx}\operatorname{cosech}(x) = -\operatorname{cosech}(x) \cdot \coth(x)$$

5. **Hyperbolic Secant**

   The derivative of the hyperbolic secant function involves both the hyperbolic secant and tangent functions.

$$\frac{d}{dx}\operatorname{sech}(x) = -\operatorname{sech}(x) \cdot \tanh(x)$$

6. **Hyperbolic Cotangent**

   The derivative of the hyperbolic cotangent function is the negative square of the hyperbolic cosecant function.

$$\frac{d}{dx}\coth(x) = -\operatorname{cosech}^2(x)$$

2.6. Derivative of Inverse Hyperbolic Functions

Inverse hyperbolic functions provide the corresponding hyperbolic angles for given values, with derivatives reflecting this relationship. These functions are used in various applications, such as integrating certain types of functions and solving inverse hyperbolic equations.

1. **Inverse Hyperbolic Sine**

   The derivative of the inverse hyperbolic sine function shows how the rate of change varies with the argument's hyperbolic value.

$$\frac{d}{dx} sinh^{-1} x = \frac{1}{\sqrt{x^2+1}}$$

2. **Inverse Hyperbolic Cosine**

   The derivative of the inverse hyperbolic cosine function reflects the relationship between the rate of change and the argument's hyperbolic value.

$$\frac{d}{dx}\cosh^{-1} x = \frac{1}{\sqrt{x^2-1}}$$

3. **Inverse Hyperbolic Tangent**

   The derivative of the inverse hyperbolic tangent function indicates how the rate of change varies with the argument's hyperbolic value.

$$\frac{d}{dx}\tanh^{-1} x = \frac{1}{1-x^2}$$

4. **Inverse Hyperbolic Cosecant**

   The derivative of the inverse hyperbolic cosecant function involves the absolute value of the argument, reflecting its domain restrictions.

$$\frac{d}{dx}\operatorname{cosech}^{-1} x = -\frac{1}{|x|\sqrt{1+x^2}}$$

5. **Inverse Hyperbolic Secant**

   The derivative of the inverse hyperbolic secant function shows the relationship between the rate of change and the argument's hyperbolic value.

$$\frac{d}{dx}\operatorname{sech}^{-1} x = -\frac{1}{x\sqrt{1-x^2}}$$

6. **Inverse Hyperbolic Cotangent**

   The derivative of the inverse hyperbolic cotangent function is similar to that of the inverse hyperbolic tangent function but involves a different range of values.

$$\frac{d}{dx}\coth^{-1} x = \frac{1}{1-x^2}$$

Differentiation of various types of functions, including trigonometric, inverse trigonometric, exponential, logarithmic, hyperbolic, and inverse hyperbolic functions, is a crucial aspect of calculus. Mastery of these differentiation techniques allows for the analysis of a wide range of mathematical problems and their applications in science, engineering, and other fields. By understanding and practicing these rules, one can develop a deeper understanding of how functions change and how to measure these changes precisely.

3. WORKING EXAMPLES

1. Differentiate $y = sin\, x$

$$\frac{dy}{dk} = \frac{d}{dx} sin\, x = cos\, x$$

2. Differentiate $y = e^x$

$$\frac{dy}{dx} = \frac{d}{dx} e^x = e^x$$

3. Differentiate $y = tan\, x$

$$\frac{dy}{dx} = \frac{d}{dx} tan\, x = sec^2\, x$$

4. Differentiate $y = ln(x^2 + 1)$

   Using the chain rule:

$$\frac{dy}{dx} = \frac{d}{dx} ln(x^2 + 1)$$

$$\frac{dy}{dx} = \frac{1}{x^2 + 1} \frac{d}{dx}(x^2 + 1)$$

$$\frac{dy}{dx} = \frac{1}{x^2 + 1} \cdot 2x = \frac{2x}{x^2 + 1}$$

5. Differentiate $y = sinh(x)$

$$\frac{dy}{dx} = \frac{d}{dx} sinh\, x = cosh\, x$$

6. Differentiate $y = x \cdot e^{x^2}$

   Using the Product and Chain Rule:

$$\frac{dy}{dx} = \frac{d}{dx} x \cdot e^{x^2}$$

$$\frac{dy}{dx} = x \cdot \frac{d}{dx} e^{x^2} + e^{x^2} \cdot \frac{d}{dx} x$$

$$\frac{dy}{dx} = x \cdot e^{x^2} \cdot \frac{d}{dx} x^2 + e^{x^2} \cdot 1$$

$$\frac{dy}{dx} = x \cdot e^{x^2} \cdot 2x + e^{x^2} \cdot 1$$

$$\frac{dy}{dx} = 2x^2 e^{x^2} + e^{x^2}$$

$$\frac{dy}{dx} = e^{x^2}(2x^2 + 1)$$

7. Differentiate $y = (x^2 + 1)^5$

   Using the Chain Rule:

$$\frac{dy}{dx} = \frac{d}{dx}(x^2 + 1)^5$$

$$\frac{dy}{dx} = 5(x^2 + 1)^4 \cdot \frac{d}{dx}(x^2 + 1)$$

$$\frac{dy}{dx} = 5(x^2 + 1)^4 \cdot 2x$$

$$\frac{dy}{dx} = 10x^{(x^2+1)^4}$$

8. Differentiate $y = \frac{sin\, x}{x^2}$

   Using the Quotient Rule:

$$\frac{dy}{dx} = \frac{\frac{d}{dx} sin\, x \cdot x^2 - sin\, x \cdot \frac{d}{dx} x^2}{(x^2)^2}$$

$$\frac{dy}{dx} = \frac{\cos x \cdot x^2 - \sin x \cdot 2x}{x^4}$$

$$\frac{dy}{dx} = \frac{x^2 \cos x - 2x \sin x}{x^4}$$

$$\frac{dy}{dx} = \frac{\cos x}{x^2} - \frac{2 \sin x}{x^2}$$

9. Differentiate $y = tan^{-1} x$

$$\frac{dy}{dx} = \frac{d}{dx} tan^{-1} x = \frac{1}{1 + x^2}$$

10. Find the second order derivative of $y = ln(x^2 + 1)$

    First, find the first derivative using the Chain Rule:

$$\frac{dy}{dx} = \frac{2x}{x^2 + 1}$$

    Now, find the second derivative using the Quotient Rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{2x}{x^2 + 1}\right)$$

$$\frac{d^2y}{dx^2} \cdot = \frac{\frac{d}{dx} 2x \cdot (x^2 + 1) - 2x \cdot \frac{d}{dx}(x^2 + 1)}{(x^2 + 1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2(x^2 + 1) - 2x \cdot 2x}{(x^2 + 1)^2}$$

$$\frac{d_y^2}{dx^2} = \frac{2x^2 + 2 - 4x^2}{(x^2+1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2 - 2x^2}{(x^2+1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2(1 - x^2)}{(x^2+1)^2}$$

4. SUMMARY

As we wrap up Unit 10 on the Differentiation of Trigonometric Functions and Inverse Circular Functions, let's take a moment to reflect on what we've covered and how it ties into the broader world of calculus and its applications.

Firstly, we dove into the derivatives of basic trigonometric functions. It's fascinating to see how each function's rate of change is intimately tied to others within the trigonometric family. For example, the derivative of the sine function is the cosine function, which really illuminates the cyclic nature of these relationships. Then, we explored the tangent function, whose derivative is the secant squared function, showing how quickly things can escalate in trigonometry due to its steep curves near vertical asymptotes.

Our journey didn't stop there; we ventured into the realm of inverse trigonometric functions. These functions, like arcsine or arctangent, are crucial for retrieving angles from known trigonometric ratios, a common task in fields ranging from engineering to physics. Understanding their derivatives helps in solving integration problems and in situations where rates of change are pivotal, such as in motion dynamics.

Moreover, the derivative processes themselves offer a toolkit for tackling complex equations, particularly when functions are composed or intertwined. This was apparent as we applied the chain rule, product rule, and quotient rule in various contexts to demonstrate the versatility and necessity of these rules in calculus.

Let's consider why mastering these differentiation techniques matters. Beyond the academic pursuit, these skills are essential for any scientific field that deals with waves, oscillations, or circular motions—areas where trigonometric functions naturally describe phenomena. For instance, engineers analyzing signal waves or economists modelling cyclical trends both rely on these mathematical principles.

To effectively learn and apply the concepts from this unit, I encourage you to practice regularly and tackle diverse problems. This not only reinforces the formulas and rules but also enhances your ability to see underlying patterns and connections, which is invaluable for deeper comprehension and innovative problem-solving.

This unit has not only equipped you with the mathematical skills to handle specific functions and their rates of change but also highlighted the interconnected nature of mathematical concepts, readying you for more complex challenges ahead. Whether you're plotting the course of a satellite or optimizing a financial model, the tools you've honed here will be part of your essential arsenal in the realm of applied mathematics.

INSPIRED BY LIFE

5. SELF-ASSESSMENT QUESTIONS

1. The derivative of $sin\, x$ is _____.
2. $\frac{d}{dx} cos\, x$ equals _____.
3. The derivative of $tan\, x$ results in _____.
4. $\frac{d}{dx} cot\, x =$_____.
5. For the function $sec\, x$, the derivative is _____.
6. The derivative of $cosec\, x$ can be expressed as _____.
7. The derivative of $sin^{-1} x$ is _____.
8. $\frac{d}{dx} cos^{-1} x =$_____.
9. $\frac{d}{dx} tan^{-1} x =$_____.
10. The derivative of $cot^{-1} x$ is _____.

6. TERMINAL QUESTIONS

1. Find the derivative of $y = 3\,sin\,x$
2. Calculate the derivative of $y = cos\,2x$
3. Determine the derivative of $y = tan\frac{\pi}{2}$
4. What is the derivative of $y = 5\,cot\,x$?
5. Compute the derivative of $y = 4\,sec\,x$
6. Compute the derivative of $y = cot^{-1}(\sqrt{x})$
7. Find the Derivative of $y = sin(x^2 + 1)$
8. Calculate the derivative of $y = cos(x + sin\,x)$
9. What is the derivative of $y = cot(x^3 - x)$
10. Calculate the derivative of $y = 2\,sin^{-1}x + 3\,cos^{-1}x$

7. ANSWERS

Self-Assessment Questions

1. $cos\, x$
2. $-\, sin\, x$
3. $\sec^2 x$
4. $-\, cosec^2\, x$
5. $\sec x \tan x$
6. $-\, cosec\, x \cdot cot\, x$
7. $\frac{1}{\sqrt{1-x^2}}$
8. $-\frac{1}{\sqrt{1-x^2}}$
9. $\frac{1}{1+x^2}$
10. $-\frac{1}{1+x^2}$

Terminal Questions

1. $3\, cos\, x$
2. $-2\, sin(2x)$
3. $\frac{1}{2}\sec^2(2x)$
4. $-5\, \mathrm{cosec}^2\, x$
5. $4 \sec x \cdot \tan x$
6. $-\frac{1}{2x\sqrt{x}}$
7. $2x \cos(x^2+1)$
8. $(1-cos\, x)\, sin(x+sin\, x)$
9. $(3x^2-1)\, cosec^2(x^3-x)$
10. $\frac{2}{\sqrt{1-x^2}} - \frac{3}{\sqrt{1-x^2}}$